

Дискретно-событийное моделирование

Лабораторная работа №7

Другие форматы

 Beamer

 RevealJS

АВТОР

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ДАТА ПУБЛИКАЦИИ

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Информация

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Вводная часть

Актуальность

- Дискретно-событийное моделирование широко применяется для анализа систем массового обслуживания
- Модели M/M/c и Росса позволяют оценивать производительность и надёжность реальных систем
- Важно уметь реализовывать такие модели программно и интерпретировать результаты

Объект и предмет исследования

- **Объект:** системы массового обслуживания
- **Предмет:** дискретно-событийное моделирование моделей M/M/c и Росса на языке Julia

Цели и задачи

- Освоить метод дискретно-событийного моделирования
- Реализовать модели M/M/c и Росса на Julia
- Провести параметрические исследования
- Построить графики и сравнить с аналитикой

Материалы и методы

- Язык программирования Julia
- Пакеты: [ConcurrentSim](#), [ResumableFunctions](#), [Distributions](#), [Plots](#), [Literate](#), [DrWatson](#)
- Quarto для генерации отчётов и презентаций

Создание презентации

Процессор [quarto](#)

- Quarto: универсальный издательский инструмент
- Поддержка Julia, Python, R
- Создание PDF, HTML, DOCX

Формат [pdf](#)

- Использование LaTeX
- Пакет для презентации: beamer
- Автоматическая компиляция через [quarto render](#)

Код для формата [pdf](#)

```
format:
  beamer:
    theme: metropolis
    aspectratio: 169
    slide-level: 2
```

Формат [html](#)

- Используется фреймворк reveal.js
- Интерактивная презентация
- Поддержка анимации и переходов

Код для формата [html](#)

```
format:
  revealjs:
    theme: beige
    slide-number: true
```

Выполнение лабораторной работы

1. Создание нового релиза

- Создан релиз v.7.0.0 для лабораторной работы №7

```

presentation report
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab07$ git flow release start
7.0.0
Branches 'develop' and 'origin/develop' have diverged.
And local branch 'develop' is ahead of 'origin/develop'.
M      labs/lab06/project
Switched to a new branch 'release/7.0.0'

Summary of actions:
- A new branch 'release/7.0.0' was created, based on 'develop'
- You are now on branch 'release/7.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

git flow release finish '7.0.0'

```

Рисунок 1

2. Настройка окружения Julia

- Запуск Julia и установка пакетов

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab07$ julia

Documentation: https://docs.julialang.org
Type "?" for help, "I?" for Pkg help.
Version 1.10.5 (2024-08-27)
Official https://julialang.org/ release

julia> using Pkg

julia> Pkg.add("DrWatson")

Updating registry at `~/\.julia/registries/General.toml`
Resolving package versions...
No Changes to `~/\.julia/environments/v1.10/Project.toml`
No Changes to `~/\.julia/environments/v1.10/Manifest.toml`

julia>

julia> using DrWatson

julia>

julia> initialize_project("project"; authors=["Алина"], git=true)
Activating new project at `~/work/study/2026-1/2026-1==study--simulati
on-modeling/2026-1--study--simulation-modeling/labs/lab07/project`
Resolving package versions...
Installed ScopedValues - v1.6.2
Installed JLLWrappers - v1.8.0
Installed FileIO - v1.19.0
Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab07/project/Project.toml`
[634d3b9d] + DrWatson v2.19.1
Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/202

```

Рисунок 2

```

[05823500] + OpenLibm_jll v0.8.1+2
[efcefd7] + PCRE2_jll v10.42.0+1
[bea87d4a] + SuiteSparse_jll v7.2.1+1
[8e850b90] + libblastrampoline_jll v5.11.0+0
Info Packages marked with ⚠ have new versions available but comp
atibility constraints restrict them from upgrading. To see why use `stat
us --outdated -m`
Precompiling project...
188 dependencies successfully precompiled in 318 seconds. 131 already
precompiled.

✅ Все пакеты установлены!
Для проверки: using DrWatson, DifferentialEquations, Plots

```

Рисунок 3

3. Установка пакетов для скриптов

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab07/project$ julia --projec
t=. -e 'using Pkg; Pkg.add("StableRNGs")'
Resolving package versions...
Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab07/project/Project.toml`
[860ef19b] + StableRNGs v1.0.4
No Changes to `~/work/study/2026-1/2026-1==study--simulation-modeling/
2026-1--study--simulation-modeling/labs/lab07/project/Manifest.toml`

```

Рисунок 4

```

scripts/mmc_model.jl:4
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab07/project$ julia --projec
t=. -e 'using Pkg; Pkg.add("Distributions")'
6-1--study--simulation-modeling/labs/lab07/project/Project.toml`
[31c24e10] + Distributions v0.25.125
Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab07/project/Manifest.toml`
[31c24e10] + Distributions v0.25.125
[1a297f60] + FillArrays v1.16.0
[34004b35] + HypergeometricFunctions v0.3.28
[90014a1f] + PDMats v0.11.37
[1fd47b50] + QuadGK v2.11.3
[79098fc4] + Rmath v0.9.0
[4c63d2b9] + StatsFuns v1.5.2
[f50d1b31] + Rmath_jll v0.5.1+0
[4607b0f0] + SuiteSparse
Precompiling project...
8 dependencies successfully precompiled in 23 seconds. 327 already pre
compiled.

```

Рисунок 5

```

(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab07/project$ julia --projec
t=. -e 'using Pkg; Pkg.add("ConcurrentSim")'
Resolving package versions...
Installed ResumableFunctions - v1.0.6
Installed ConcurrentSim v1.5.1
Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab07/project/Project.toml`
[6ed1e86c] + ConcurrentSim v1.5.1
Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab07/project/Manifest.toml`
[6ed1e86c] + ConcurrentSim v1.5.1
[c5292f4c] + ResumableFunctions v1.0.6
Precompiling project...
2 dependencies successfully precompiled in 13 seconds. 335 already pre
compiled.

```

Рисунок 6

```

scripts/mmc_model.jl:4
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab07/project$ julia --projec
t=. -e 'using Pkg; Pkg.add("ResumableFunctions")'
Resolving package versions...
Updating `~/work/study/2026-1/2026-1==study--simulation-modeling/202
6-1--study--simulation-modeling/labs/lab07/project/Project.toml`
[c5292f4c] + ResumableFunctions v1.0.6
No Changes to `~/work/study/2026-1/2026-1==study--simulation-modeling/
2026-1--study--simulation-modeling/labs/lab07/project/Manifest.toml`
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli

```

Рисунок 7

4. Запуск модели M/M/c

- Запуск скрипта `scripts/mmc_model.jl`

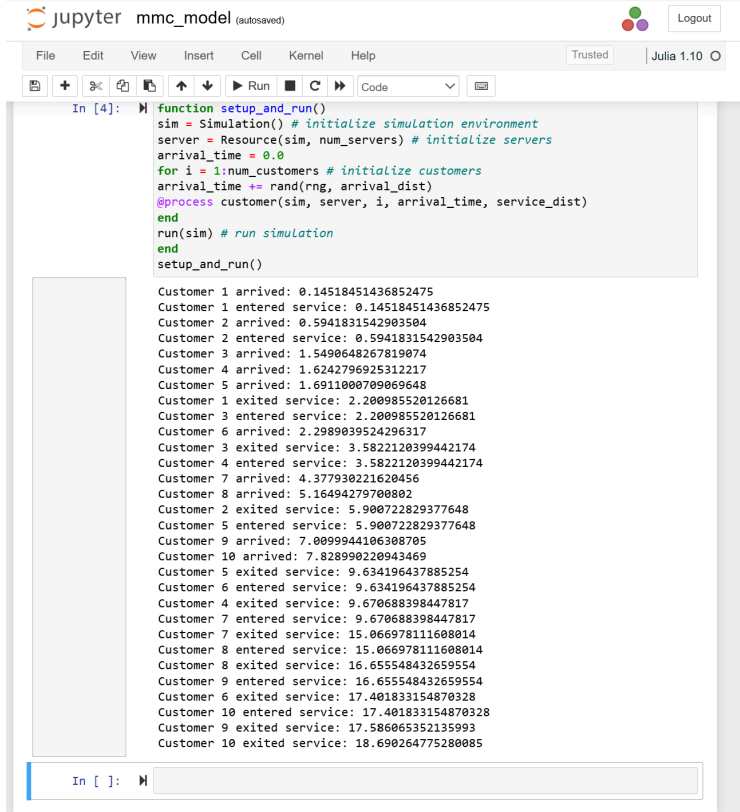
```

2026-1--study--simulation-modeling/labs/lab07/project/Manifest.toml
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeli
ng/2026-1--study--simulation-modeling/labs/lab07/project$ julia --projec
t=. scripts/mmc_model.jl
Customer 1 arrived: 0.14518451436852475
Customer 1 entered service: 0.14518451436852475
Customer 2 arrived: 0.5941831542903504
Customer 2 entered service: 0.5941831542903504
Customer 3 arrived: 1.5490648267819074
Customer 4 arrived: 1.6242796925312217
Customer 5 arrived: 1.6911000709069648
Customer 1 exited service: 2.200985520126681
Customer 3 entered service: 2.200985520126681
Customer 6 arrived: 2.2989039524296317
Customer 3 exited service: 3.5822120399442174
Customer 4 entered service: 3.5822120399442174
Customer 7 arrived: 4.377930221620456
Customer 8 arrived: 5.16494279700802
Customer 2 exited service: 5.900722829377648
Customer 5 entered service: 5.900722829377648
Customer 9 arrived: 7.0099944106308705
Customer 10 arrived: 7.828990220943469
Customer 5 exited service: 9.634196437885254
Customer 6 entered service: 9.634196437885254
Customer 4 exited service: 9.670688398447817
Customer 7 entered service: 9.670688398447817
Customer 7 exited service: 15.066978111608014
Customer 8 entered service: 15.066978111608014
Customer 8 exited service: 16.655548432659554
Customer 9 entered service: 16.655548432659554
Customer 6 exited service: 17.401833154870328
Customer 10 entered service: 17.401833154870328
Customer 9 exited service: 17.586065352135993
Customer 10 exited service: 18.690264775280085

```

Рисунок 8

- Создан Jupyter Notebook для модели M/M/c



The screenshot shows a Jupyter Notebook window titled "mms_model (autosaved)". The interface includes a menu bar (File, Edit, View, Insert, Cell, Kernel, Help), a toolbar with icons for file operations and execution, and a status bar at the bottom indicating "Julia 1.10". The notebook contains a single code cell with the following Julia code:

```
In [4]: function setup_and_run()
        sim = Simulation() # initialize simulation environment
        server = Resource(sim, num_servers) # initialize servers
        arrival_time = 0.0
        for i = 1:num_customers # initialize customers
            arrival_time += rand(rng, arrival_dist)
            @process customer(sim, server, i, arrival_time, service_dist)
        end
        run(sim) # run simulation
    end
    setup_and_run()
```

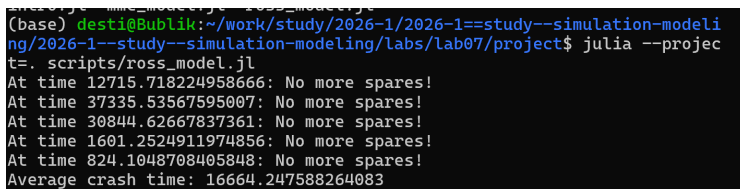
The output of the code cell displays a log of simulation events for 10 customers, including arrival and service times:

```
Customer 1 arrived: 0.14518451436852475
Customer 1 entered service: 0.14518451436852475
Customer 2 arrived: 0.5941831542903504
Customer 2 entered service: 0.5941831542903504
Customer 3 arrived: 1.5490648267819074
Customer 4 arrived: 1.6242796925312217
Customer 5 arrived: 1.6911000709069648
Customer 1 exited service: 2.200985520126681
Customer 3 entered service: 2.200985520126681
Customer 6 arrived: 2.2989039524296317
Customer 3 exited service: 3.5822120399442174
Customer 4 entered service: 3.5822120399442174
Customer 7 arrived: 4.377930221620456
Customer 8 arrived: 5.16494279700802
Customer 2 exited service: 5.900722829377648
Customer 5 entered service: 5.900722829377648
Customer 9 arrived: 7.0099944106308705
Customer 10 arrived: 7.828990220943469
Customer 5 exited service: 9.634196437885254
Customer 6 entered service: 9.634196437885254
Customer 4 exited service: 9.670688398447817
Customer 7 entered service: 9.670688398447817
Customer 7 exited service: 15.066978111608014
Customer 8 entered service: 15.066978111608014
Customer 8 exited service: 16.65548432659554
Customer 9 entered service: 16.65548432659554
Customer 6 exited service: 17.401833154870328
Customer 10 entered service: 17.401833154870328
Customer 9 exited service: 17.586065352135993
Customer 10 exited service: 18.690264775280085
```

Рисунок 9

5. Запуск модели Росса

- Запуск скрипта `scripts/ross_model.jl`



The screenshot shows a terminal window with the following text:

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project$ julia --project=. scripts/ross_model.jl
At time 12715.718224958666: No more spares!
At time 37335.53567595007: No more spares!
At time 30844.62667837361: No more spares!
At time 1601.2524911974856: No more spares!
At time 824.1048708405848: No more spares!
Average crash time: 16664.247588264083
```

Рисунок 10

- Создан Jupyter Notebook для модели Росса

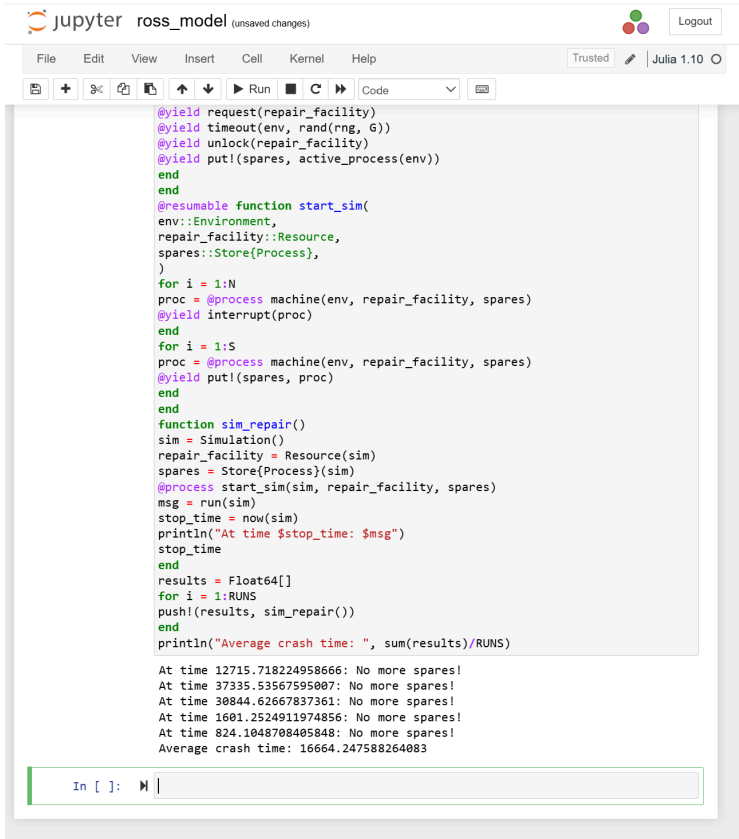


Рисунок 11

6. Создание производных форматов

- Генерация чистого кода, Notebook и Quarto-документов

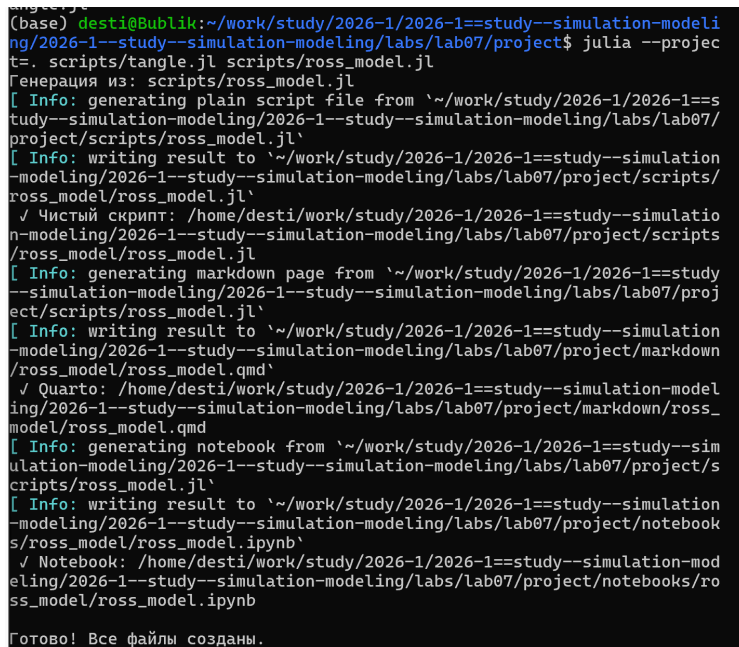


Рисунок 12

```
(base) desti@Bublik:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project$ julia --project=. scripts/tangle.jl scripts/mmc_model.jl
Генерация из: scripts/mmc_model.jl
[ Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project/scripts/mmc_model.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project/scripts/mmc_model/mmc_model.jl`
✓ Чистый скрипт: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project/scripts/mmc_model/mmc_model.jl
[ Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project/scripts/mmc_model.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project/markdown/mmc_model/mmc_model.qmd`
✓ Quarto: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project/markdown/mmc_model/mmc_model.qmd
[ Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project/scripts/mmc_model.jl`
[ Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project/notebooks/mmc_model/mmc_model.ipynb`
✓ Notebook: /home/desti/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab07/project/notebooks/mmc_model/mmc_model.ipynb

Готово! Все файлы созданы.
```

Рисунок 13

7. Настройка отчёта

- В `_quarto.yml` добавлена поддержка Julia

```
!_quarto.yml ×
Ubuntu > home > desti > work > study > 2026-1 > 2026-1==study--simulation-mo

1 project:
2   title: "report"
3   output-dir: "_output"
4
5 standalone: true
6 self-contained: true
7
8 ## Julia support
9 # engine: julia
10 # julia:
11 #   exeflags: ["--project=../project"]
12
13 ## Generic options
14 lang: ru-RU
15 number-sections: true
16 toc: true
17 toc-title: "Содержание"
18 toc-depth: 2
19 ## Crossref customization
20 crossref:
21   lof-title: "Список иллюстраций"
22   lot-title: "Список таблиц"
23   lol-title: "Листинги"
24 ## Bibliography
25 bibliography:
26   - bib/cite.bib
27   csl: _resources/csl/gost-r-7-0-5-2008-numeric.csl
28 ## Formats
29 format:
30   ### Docx output format
31   docx:
32     toc: true
33     number-sections: true
34     toc-depth: 2
35   ### Pdf output format
36   pdf:
37     # pdf-engine: xelatex
38     ### minted support
39     # filters:
40     #   - minted-quarto
41     toc: true
42     number-sections: true
43     colorlinks: false
44     toc-depth: 2
45     lof: true # List of figures
46     lot: true # List of tables
47     ##### Document
48     documentclass: scrreprt
49     papersize: a4
50     fontsize: 12pt
51     linestretch: 1.5
52     ##### Language
53     babel-lang: russian
54     babel-otherlangs: english
55     ##### Biblatex
56     cite-method: biblatex
57     biblio-style: gost-numeric
58     biblatexoptions:
59       - backend=biber
60       - langhook=extras
61       - autolang=other*
62     ##### Misc options
63     csquotes: true
64     indent: true
65     include-in-header:
66       - file: _resources/tex/preamble.tex
67
```

Рисунок 14

8. Подключение документации

- В отчёт подключены сгенерированные .qmd файлы

```
{{< include ../project/markdown/mmc_model/mmc_model.qmd >}}  
{{< include ../project/markdown/ross_model/ross_model.qmd >}}
```

Рисунок 15

```
```{julia}  
using StableRNGs
using Distributions
using ConcurrentSim
using ResumableFunctions
#set simulation parameters
rng = StableRNG(123)
num_customers = 10 # total number of customers generated
```

set queue parameters

```
num_servers = 2 # number of servers
mu = 1.0 / 2 # service rate
lam = 0.9 # arrival rate
arrival_dist = Exponential(1 / lam) # interarrival time distribution
service_dist = Exponential(1 / mu) # service time distribution
```

define customer behavior

```
@resumable function customer(
 env::Environment,
 server::Resource,
 id::Integer,
 t_a::Float64,
 d_s::Distribution,
)
@yield timeout(env, t_a) # customer arrives
println("Customer $id arrived: ", now(env))
@yield request(server) # customer starts service
println("Customer $id entered service: ", now(env))
@yield timeout(env, rand(rng, d_s)) # server is busy
@yield unlock(server) # customer exits service
println("Customer $id exited service: ", now(env))
end
```

setup and run simulation

```
function setup_and_run()
 sim = Simulation() # initialize simulation environment
 server = Resource(sim, num_servers) # initialize servers
 arrival_time = 0.0
 for i = 1:num_customers # initialize customers
 arrival_time += rand(rng, arrival_dist)
 @process customer(sim, server, i, arrival_time, service_dist)
 end
 run(sim) # run simulation
end
setup_and_run()

using ResumableFunctions
using ConcurrentSim
using Distributions
```

```

using Random
using StableRNGs
const RUNS = 5
const N = 10
const S = 3
const SEED = 150
const LAMBDA = 100
const MU = 1
const rng = StableRNG(42) # setting a random seed for reproducibility
const F = Exponential(LAMBDA)
const G = Exponential(MU)
@resumable function machine(
 env::Environment,
 repair_facility::Resource,
 spares::Store{Process},
)
 while true
 try
 @yield timeout(env, Inf)
 catch
 end
 @yield timeout(env, rand(rng, F))
 get_spare = take!(spares)
 @yield get_spare | timeout(env)
 if state(get_spare) != ConcurrentSim.idle
 @yield interrupt(value(get_spare))
 else
 throw(StopSimulation("No more spares!"))
 end
 @yield request(repair_facility)
 @yield timeout(env, rand(rng, G))
 @yield unlock(repair_facility)
 @yield put!(spares, active_process(env))
 end
end

@resumable function start_sim(
 env::Environment,
 repair_facility::Resource,
 spares::Store{Process},
)
 for i = 1:N
 proc = @process machine(env, repair_facility, spares)
 @yield interrupt(proc)
 end
 for i = 1:S
 proc = @process machine(env, repair_facility, spares)
 @yield put!(spares, proc)
 end
end

function sim_repair()
 sim = Simulation()
 repair_facility = Resource(sim)
 spares = Store{Process}(sim)
 @process start_sim(sim, repair_facility, spares)
 msg = run(sim)
 stop_time = now(sim)
 println("At time $stop_time: $msg")
 stop_time
end

```

```
end
results = Float64[]
for i = 1:RUNS
push!(results, sim_repair())
end
println("Average crash time: ", sum(results)/RUNS)

'''
```

## Результаты

### Реализованные модели

---

- Реализованы обе модели на Julia с использованием [ConcurrentSim](#)
- Код оформлен в литературном стиле
- Сгенерированы чистые скрипты, Notebook и Quarto-документы

### Параметрические исследования

---

- **М/М/с:** исследовано влияние числа каналов на длину очереди и время ожидания
- **Модель Росса:** исследовано влияние числа ремонтников и резервных машин на среднее время до падения системы

### Построенные графики

---

- Динамика числа заявок в системе
- Загрузка ремонтника
- Изменение числа исправных машин во времени

### Итоговый отчёт

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- Отчёт скомпилирован в PDF
- Все результаты интегрированы в единый документ

## Итоговый слайд

### Основной вывод

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- Дискретно-событийное моделирование — эффективный инструмент анализа СМО
- Библиотека [ConcurrentSim](#) в Julia позволяет гибко реализовывать модели любой сложности
- Литературное программирование обеспечивает воспроизводимость и прозрачность исследований

# Рекомендую

## Принцип 10/20/30

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- 10 слайдов
- 20 минут на доклад
- 30 кегль шрифта

## Связь слайдов

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- Один слайд — одна мысль
- Нельзя ссылаться на объекты с предыдущих слайдов
- Каждый слайд должен иметь заголовок

## Количество сущностей

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- Человек может одновременно помнить  $7 \pm 2$  элемента
- На слайде — не более 5 элементов
- Можно группировать элементы до 5 групп

## Общие рекомендации

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- Слайды дополняют выступление, а не дублируют его
- Информация изложена кратко и структурированно
- Слайд не перегружен графикой и текстом
- Минимум анимации и переходов

## Представление данных

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- Лучше — в виде схемы
- Менее оптимально — рисунок, график, таблица
- Текст — только если другие способы не подошли

## Список литературы

- Имитационное моделирование. Практикум.  
<https://esystem.rudn.ru/pluginfile.php/3094278/modeling-lab.pdf>

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Повторное использование

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